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Enhanced Correlation Mapping Framework Employing Hybrid Evolutionary Strategy for Drug–Disease Association Discovery Across Diverse Biomedical Sources

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ABSTRACT: Hepatocellular carcinoma (HCC) is one of the most common malignant tumors with a high rate of morbidity and mortality. Predicting new drug indications in drug development for HCC analysis becomes a major and difficult task. Few algorithms have been devoted to drug prediction for a specific disease. This work provides an integrative framework to predict novel drugs and diseases for HCC based on a multi-source Imperialist Competitive Algorithm-based random walk (PDDH-MICARW). Firstly, based on gene expression and disease interaction network, construct a gene-gene weighted interaction network (GWIN). Then, dimensionality reduction is performed by using the Improved Regularized Linear Discriminant Analysis (IRLDA). Then, based on a multi-source random walk in GWIN, build a drug-drug similarity network and, disease-drug similarity network. Finally, based on the known drugs for HCC, score all drugs in the drug-drug similarity network, disease-drug bipartite graph network. This network provides more information on these interactions, compared to the component or parent networks. The robustness of PDDH-MICARW predictions is measured for HCC. It concludes that the proposed work can effectively identify new drug indications. The robustness of our predictions, their overlap with those reported in the Comparative Toxicogenomics Database (CTD) and literature, and their enriched Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway demonstrate proposed approach can effectively identify new drug indications.

KEYWORDS: Hepatocellular Carcinoma (HCC), dimensionality reduction, Improved Regularized Linear Discriminant Analysis (IRLDA), gene-Gene Weighted Interaction Network (GWIN), and Multi-source Imperialist Competitive Algorithm-based Random Walk (MICARW).

1 Introduction

Nature presented itself as a potential drug resource for scientists, enabling serendipitous discoveries for centuries. The venture of drug discovery started with random screening of crude plants such as reserpine, lavender, and opium, and later created the drug industry, which is one of the most advanced industries in our society [1]. As a part of this industry, the drug discovery process has enormous investments in infrastructure. Research and development (R&D) require huge amounts of time and money. Further, translation of the promising molecule to an approved drug has



major problems, including high failure rates and withdrawal risks by reason of safety or efficacy problems [2].

Drug repositioning (DR), which is an auxiliary process to conventional drug discovery, has gained attention from both academia and pharmaceutical companies. Drug repositioning, also named “drug repurposing or drug recycling,” is a promising approach to overcome hurdles in discovering and developing new drugs via the identification of new therapeutic applications of known drugs. Repurposed drug candidates have established formulations and manufacturing methods, extensive pharmacokinetic properties, known adverse effects, clinical trial information, and post-marketing surveillance safety data, which make a beeline for the drug development with less chance of failure [3]. With the generation of large-scale genomic, transcriptomic, and proteomic data, computational prediction of new drug indications has become one of the leading ways to investigate drug-disease treatment. Several computational methods have been introduced to solve this problem. Those methods are discussed in detail in the review section.

In the review work few methods have been introduced to solve the drug prediction for a specific disease. This is also one reason why the prediction accuracy of many results is not high enough. Hepatocellular Carcinoma (HCC) is the sixth most common cancer and the third leading cause of cancer-related death worldwide, and its incidence is increasing [4]. Despite the widespread use of surveillance programs in at-risk populations, more than half of HCC cases are diagnosed late, and curative therapies such as surgical resection, transplantation, or radiofrequency ablation are possible in fewer than 30% of the patients. Most HCC patients have unresectable disease at diagnosis [5]. The prognosis of patients with HCC remains poor, and life expectancy is difficult to predict. HCC development is often related to the presence of a chronic liver disease, which represents one of the most important risk factors for this neoplasia. In particular, cirrhosis, which can occur as a consequence of chronic viral hepatitis, excessive alcohol intake, nonalcoholic fatty liver disease, or genetic diseases (e.g., hemochromatosis), is a frequent setting for HCC onset as well as a cause of liver dysfunction.

In this study, the precise drug prediction is taken as an extremely hard process. As a result, this research is used to find out numerous accurate drug signs for HCC. This method finds novel drugs and diseases for HCC based on predict novel drugs and diseases for HCC based on multi-source Imperialist Competitive Algorithm-based random walk (PDDH-MICARW) with several steps: at the start of the research, integrate gene expression data, a gene set, and a drug disease similarity network to construct a GWIN for HCC. Subsequently, the data dimensionality is decreased by means of utilizing Improved Regularized Linear Discriminant Analysis (IRLDA) to enhance the prediction accuracy. After that, based on GWIN, a disease-drug similarity network and a drug-drug similarity network are constructed. Finally, taking all the existing drugs for the treatment of HCC, all drugs in the networks are scored to find new appropriate drugs for HCC.

2 Literature Review

Li et al [6] developed a computational framework to build disease-specific drug-protein connectivity maps. Also integrated gene/protein and drug connectivity information based on protein interaction networks and literature mining, without requiring gene expression profile information derived from drug perturbation experiments on disease samples. To achieve overall good data coverage and quality, a series of statistical methods has been developed to overcome high levels of data noise in biological networks and literature mining results.

Dudley et al [7] showed that computational methods for drug repositioning can be classified in two axes: drug-based, where discovery initiates from the chemical perspective, or disease-based, where discovery initiates from the clinical perspective of disease or its pathology. Newer algorithms for computational drug repositioning will likely span these two axes, will take advantage of newer

types of molecular measurements, and will certainly play a role in reducing the global burden of disease.

Jin et al [8] defined a new network component called Cancer-Signaling Bridges (CSB) and integrated it with a Bayesian Factor Regression Model (BFRM) to form a new hybrid method termed CSB-BFRM. Proof-of-concept studies were conducted in breast and prostate cancer cells and in promyelocytic leukemia cells. In each system, CSB-BFRM analysis could accurately predict clinical responses to more than 90% of drugs approved by the U.S. Food and Drug Administration and more than 75% of experimental clinical drugs that were tested. Mechanistic investigation of Off-Target Effects (OTE) for several high-ranking drug-dose pairs suggested repositioning opportunities for cancer therapy, based on the ability to enforce retinoblastoma-dependent repression of important E2F-dependent cell-cycle genes.

Hurle et al [9] reviewed highlights of computational techniques for systematic analysis of transcriptomics, such as Connectivity Map (CMap), side effects, and genetics Genome-Wide Association Study (GWAS) data to generate new hypotheses for additional indications. Also discuss data domains such as Electronic Health Records (EHRs) and phenotypic screening that we consider promising for novel computational repositioning methods.

Ye et al [10] developed a drug-drug network that was constructed based on the similarities between their clinical side effects. The indications of a drug may be inferred by the enriched FDA-approved functions of its neighbouring drugs in the network. Furthermore, the method could be extended to predict drugs not covered in the network. In summary, a method that is based on the indications enriched by network neighbors may provide new clues for drug repositioning using side effects.

Wang et al [11] proposed a computational framework based on a heterogeneous network model and applied the approach to drug repositioning by using existing omics data about diseases, drugs, and drug targets. The novelty of the framework lies in the fact that the strength between a disease-drug pair is calculated through an iterative algorithm on the heterogeneous graph that also incorporates drug-target information. Comprehensive experimental results show that the proposed approach significantly outperforms several recent approaches. Case studies further illustrate its practical usefulness.

Yang et al [12] proposed a Causal Inference-Probabilistic Matrix Factorization (CI-PMF) approach was proposed to predict and classify drug-disease associations, and further used for drug-repositioning predictions. In terms of drug-disease association prediction, modified causal inference included in CI-PMF outperformed existing causal inference with a higher AUC (area under the receiver operating characteristic curve) score and greater precision. Overall, CI-PMF is a useful approach for associating drugs with complex diseases and provides potential value for drug repositioning.

Jiao et al [13] provided a brief overview of recent advances in the computational drug repositioning for anticancer applications with a specific emphasis on the repositioning of non-cancer drugs by use of various computational approaches. Wang et al [14] proposed a novel recommendation-based drug repositioning strategy to simultaneously determine novel drug indications and side effects in one integrated framework. This strategy provides a complementary method to medical genetics-based drug repositioning, which reduces the occurrence of false positives in medical genetics-based drug repositioning, resulting in a ranked list of new candidate indications and/or side effects with different confidence levels. Several new drug indications and side effects are reported with high prediction confidence.

Gayvert et al [15] developed a Computational Drug Repositioning Approach for Targeting Oncogenic Transcription Factors (CRAFTT) approach for targeting TF activity. Dexamethasone

significantly decreased cell invasion and migration in an ERG-dependent manner. Furthermore, analysis of electronic medical record data indicates a protective role for dexamethasone against prostate cancer. Altogether, the method provides a broadly applicable strategy for identifying drugs that specifically modulate TF activity.

Sun et al [16] introduced a new data fusion model based on n-cluster editing as a novel multi-source triangulation strategy, which was further combined with semantic literature mining. The proposed evaluation suggests that utilizing drug-gene-disease triangulation coupled to sophisticated text analysis is a robust approach for identifying new drug candidates for repurposing.

Vanhaelen et al [17] provided a comprehensive overview of the current status of in silico repurposing methods by establishing links between current technological trends, data availability, and characteristics of the algorithms used in these methods. Using the case of the computational repurposing of fasudil as an alternative autophagy enhancer, suggest a generic modular organization of a repurposing workflow. Reviewed a three-dimensional (3D) structure-based, similarity-based, inference-based, and Machine Learning (ML)-based methods. Finish by discussing current directions of research, including possibilities offered by new deep learning.

3 Proposed Methodology

This research is used to find out numerous accurate drug signs for HCC. This method finds novel drugs and diseases for HCC based on predict novel drugs and diseases for HCC based on multi-source Imperialist Competitive Algorithm-based random walk (PDDH-MICARW) with several steps: at the start of the research, integrate gene expression data, a gene set, and a drug disease similarity network to construct a GWIN for HCC. Afterwards, the data dimensionality is decreased by means of utilizing Improved Regularized Linear Discriminant Analysis (IRLDA) to enhance the prediction accuracy. After that, based on GWIN, a disease-drug similarity network and a drug-drug similarity network are constructed. Also, with the intention of evaluating and filtering the drug-drug network, disease-drug similarity network, use z-score and a cutoff. Finally, taking the available drugs utilized for handling HCC in the form of drug seeds, all drugs in the networks are scored to find new potential drugs for HCC. As well, segregate the drugs identified by the features of their targets and get nine more reliable drugs.

3.1 Construction of Gene-Gene Weighted Interaction Network (GWIN)

The general genes along with their associated edges in the gene network are discussed in this segment. For each gene pair 'i' and 'j', while there is a relationship amongst them, calculate the Pearson Correlation Coefficient (PCC) along with its corresponding p-value. The PCC assesses the edge between gene 'i' and 'j'; instead, while its p-value ≥ 0.05 , eliminate the edge. GWIN is represented as $G = (V, E)$, wherein V and E denote the group of nodes as well as edges, respectively.

3.2 Dimensionality reduction by using IRLDA

In this section, describe the basic notations used in the work for dimension reduction of gene disease samples. Let $X = \{x_1, x_2, \dots, x_n\}$ denote n training samples (or vectors of drug disease samples) in a d-dimensional space having class labels $\Omega = \{\omega_1, \omega_2, \dots, \omega_n\}$, where $\omega \in \{1, 2, \dots, c\}$ and c is the number of classes. The gene drug diseases dataset X can be subdivided into c subsets $X_1, X_2, \dots, X_c$, where X_j belongs to class j and consists of n_j number of samples such that $n = \sum_{j=1}^c n_j$. The data subset $X_j \subset X$ and $X_1 \cup X_2 \cup \dots \cup X_c = X$. If $\mu_j = \frac{1}{n_j} \sum_{x \in X_j} x$ is the centroid of X_j and $\mu = \frac{1}{n} \sum_{x \in X} x$ is the centroid of X, then the total scatter matrix S_T , within-class scatter matrix S_W , and between-class scatter matrix S_B are defined as follows.

$$S_T = \sum_{x \in X} (x - \mu)(x - \mu)^T \quad (1)$$

$$S_W = \sum_{j=1}^c \sum_{x \in X_j} (x - \mu_j)(x - \mu_j)^T \quad (2)$$

$$S_B = \sum_{j=1}^c n_j (\mu_j - \mu)(\mu_j - \mu)^T \quad (3)$$

In the SSS problem, $d > n$, which will make scatter matrices singular. Let r_t be the rank of the S_T matrix. The eigenvector decomposition of S_T can be given as

$$S_T = [U_1, U_2] \begin{bmatrix} \Lambda_T & \cdot \\ \cdot & 0 \end{bmatrix} \begin{bmatrix} U_1^T \\ U_2^T \end{bmatrix} \quad (4)$$

where $U_1 \in \mathbb{R}^{d \times r_t}$ corresponds to eigenvalues Λ_T and $U_2 \in \mathbb{R}^{d \times (d-r_t)}$ corresponds to the zero eigenvalues. The matrix U_1 is the range space of S_T and the matrix U_2 is the null space of S_T . Since the null space of S_T does not contain any discriminant information, the dimensionality can be reduced from d -dimensional space to r_t -dimensional space by applying Principal Component Analysis (PCA) [18] as a pre-processing step. The range space of S_T matrix, $U_1 \in \mathbb{R}^{d \times r_t}$, will be used as a transformation matrix. In the reduced-dimensional space, the scatter matrices can be computed by: $S_W \leftarrow U_1^T S_W U_1$ and $S_B \leftarrow U_1^T S_B U_1$. After this procedure $S_W \in \mathbb{R}^{r_t \times r_t}$ and $S_B \in \mathbb{R}^{r_t \times r_t}$ are reduced-dimensional within-class scatter matrix and reduced-dimensional between-class scatter matrix, respectively. In RLDA, the regularization of the within-class scatter matrix S_W is carried out by adding a perturbation term α to the diagonal elements of S_W ; i.e., $\hat{S}_W = S_W + \alpha I$. The addition of α will make the within-class scatter non-singular and invertible. This would help to maximize the modified Fisher's criterion.

$$J(w, \alpha) = \frac{w^T S_B w}{w^T (S_W + \alpha I) w} \quad (5)$$

where $w \in \mathbb{R}^{r_t \times 1}$ is the orientation vector. To avoid any heuristic approach in the determination of the parameter α , we solve Eq.(5) in the following manner. Let us denote the function $f = w^T w S_B w$ and a constraint function.

$$g = w^T (S_W + \alpha I) w - c = 0 \quad (6)$$

where $c > 0$ is any constant.

3.3 Construction of drug-drug similarity network

For each drug, obtain a set of gene targets. To construct the drug-drug similarity network to be conscious of the association among all of the drugs. The association between any two drugs is derived by calculating the correlation between their target sets. By imitating the process, which involves multiple random walkers [19] with Imperialist Competitive Algorithm (ICA) called MICARW wander on GWIN, the power of association amongst any two target sets is to be scored. Figure 1 shows the flowchart of the imperialist competitive algorithm. ICA starts with an initial population of drug information. The best correlation value between the target set (countries) is selected as imperialist countries, and the rest form the imperialist colonies. All colonies of the initial population are divided among the imperialist countries based on their weight matrix of GWIN. After dividing all the colonies among the imperialists, the colonies begin the movement towards their imperialist countries. The total weight matrix of GWIN of all the empires depends on the weight matrix of GWIN of the imperialist countries and the weight matrix of GWIN of their colonies. This fact is defined with the

total weight matrix of GWIN of an empire by the weight matrix of GWIN of the imperialist country plus a percentage of the average weight matrix of GWIN of their colonies. When the imperialist competition between the empires starts, any empire that cannot remain in this competition and cannot augment its weight matrix of GWIN will be removed from the imperialist competition. The imperialist competition will result in an increment in the weight matrix of GWIN of the most weight matrix of GWIN empires and a decrement in the weight matrix of GWIN of the weaker empires. The weakest empires will lose their weight matrix of GWIN partially and, ultimately, will collapse one by one. All countries will eventually become a state in which there is only one empire in the entire world, and all remaining countries are colonies of that empire.

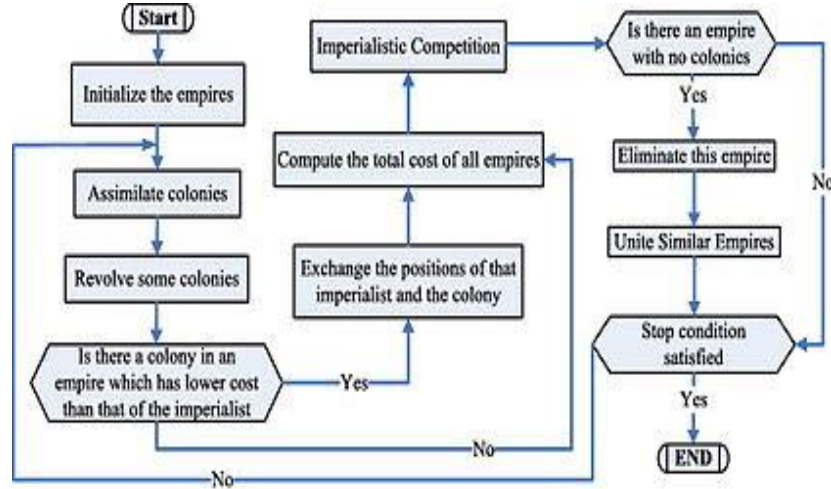


Figure 1: Flowchart of Imperialist Competitive Algorithm (ICA)

The algorithm continues with the mentioned steps (Assimilation, Revolution, Competition) until a stop condition is satisfied. The equations for the formation of empires (Initialization) are the following:

$$\text{Country} = [p_1, \dots, p_{N_{\text{var}}}] \quad (7)$$

$$\text{cost} = f(\text{country}) = f(p_1, \dots, p_n) \quad (8)$$

$$N_{\text{col}} = N_{\text{pop}} - N_{\text{imp}} \quad (9)$$

$$C_n = \max_i \{c_i\} - c_n \quad (10)$$

$$p_n = \frac{C_n}{\sum_{i=1}^{N_{\text{imp}}} C_i} \quad (11)$$

$$NC_n = \text{Round} \{p_n, N_{\text{col}}\} \quad (12)$$

In Equation (7), Country represents a country, N_{var} is the number of drugs of interest, and p_i is the value of i -th drug. Equation (8) represents the cost of a country, Equation (9) is necessary to obtain the number of colonies of the population, and used to obtain the normalized cost of each imperialist, where c_n is the n -th imperialist's cost, and C_n is the normalized cost of n -th imperialist, Equation (10)

obtains the power of each imperialist where p_n is the power of n^{th} imperialist and Equation (11) represents the number of colonies that can possess that imperialist. The equations for moving the colonies towards their imperialist country (assimilation) are:

$$x \sim U(0, \beta d) \quad (13)$$

$$\theta \sim U(-\gamma, \gamma) \quad (14)$$

Equation (13) represents a colony that moves a distance x towards its imperialist, where β is a number between 1 and 2, and d represents the distance between the drug and diseases among the colony and its imperialist. Equation (14) helps us to search for different positions around the imperialist. In which γ is a parameter, where a large value estimates a global search of the weight matrix and a small value impacts the local search of the weight matrix. The Equation for the full weight matrix of an empire is as follows:

$$TC_n = \text{Cost}(\text{imp}) + \xi \text{mean}\{\text{cost}(\text{col})\} \quad (15)$$

This equation represents the total cost of n^{th} empire, and ξ is a number between 0 and 1. Where, with a small value of ξ , it has a greater influence of the imperialist weight matrix on determining the total weight matrix of the empire, and with a large value of ξ , it has a greater influence on the average weight matrix of the colonies to determine the total weight matrix of the empire. The equations for the imperialist competition are the following:

$$NTC_n = \max_i \{TC_i\} - TC_n \quad (16)$$

$$p_{p_n} = \frac{NTC_n}{\sum_{i=1}^{N_{\text{imp}}} NTC_i} \text{ where } \sum_{i=1}^{N_{\text{imp}}} p_{p_i} = 1 \quad (17)$$

Equation (16) starts with the imperialist competition, where the normalized total cost is calculated, and in Equation (17), we provide the probability of possessing a colony. Let $S = (s_{ij})_{m \times m}$ be called the weight matrix of GWIN, which is also a transition matrix, where m is the number of genes. Let $d(0) = (d_i(0))_{m \times 1}$ is called initial probabilities. Obtain S_{ij} by allotting probabilities comparative to gene similarities with the neighbors of the complete genes in the target set TA and 0 otherwise. Let $p(0) = sd(0)$, wherein s is called an invariant for ease of computation $p^{(t)}$ includes the scores that the walkers stay at each node at time t , and comprises the iterative formula.

$$p^{t+1} = (1 - r)D^T p^{(t)} + r p^{(0)} \quad (18)$$

By iterating the iteration many steps, even now, $p^{(t)}$ is constant, and obtain the steady state $p^{(t)}$. The random walk model is not sensitive to the parameters. As a result, set default values for the parameters as $r=0$, as well as $s=1000$. Two weighted edges are there with various directions among them for each drug-drug pair. The network is represented as $G_{dd} = (V_{dd}, E_{dd})$, wherein V_{dd} and E_{dd} are similarly nodes as well as edge sets in G_{dd} .

3.4 Disease-Drug Similarity Network by using a Bipartite Graph Network

The disease drug-associated genes interconnecting with one another suggest that they tend to get grouped in the same neighborhood of the interactome and form a disease module or a drug module, a connected subgraph that comprises the entire molecular mechanisms of a disease or a

drug. As a result, the accurate evaluation of relations among disease modules as well as drug modules is considered to be an important step to identify potent drug-disease associations. A bipartite graph [20] is called a graph of two genes X and Y wherein edges (assume undirected) are simply let from one node in X to one node in Y . Consider associating a weight w with each edge in the bipartite graph, get a weighted bipartite graph. In a projection of a bipartite graph, simply one group of genes (X or Y) appears with edges from one node to another, while they were associated with the same node in the original graph. Also, the projection comprises several weights on the edges. For example, one possible weighting for an edge from node x_1 to x_2 in a projection of the set X is called the number of nodes in Y that x_1 and x_2 are associated with each other.

A graph is represented as $G = (V, E)$, wherein G is called the graph, V is known as the collection of vertices, and E is called the collection of edges. An edge connecting the vertices $u, v \in V$ is represented as $\{u, v\} \in E$. The reality that two vertices u and v are linked would also be represented as $u \sim v$. The number of neighbors of a node u is known as the degree of u and is signified as $d(u) = |\{v \in V \mid u \sim v\}|$.

Bipartite graphs are graphs wherein the collection of nodes (X, Y) is split into two separate sets, with the intention that every edge links a vertex of one partition with a vertex of the other partition. A bipartite graph is signified as

$$G = (V_1, V_2, E) \quad (19)$$

Wherein V_1 and V_2 are known as the two disjoint vertex sets, and E is called the collection of edges. A corresponding description of bipartite graphs is the graphs that do not comprise odd cycles, that is to say, cycles encompassing an odd number of edges. Especially, it keeps an eye on whether a bipartite network does not encompass triangles. Finally, in order to make the distances score comparable to the drug-disease relationships, the distance scores are treated as so. Firstly, transform all of the distances into positive by comprising the least distance score to each distance, and afterwards get their reciprocals. Finally, with the intention of normalizing the complete distances, use the maximum-minimum. As a result, the higher the distance score, the more related among drug and disease. Finally, for every disease-drug association, get its corresponding p-value. Eliminate the complete edges whose p-values are not lower than 0.01.

4 Results and Discussion

In this segment, the presented PDDH-MICARW, PDD-MSBARW-based drug prediction for HCC performance is assessed, and the performance outcomes are matched up with previous PDD with Multi-Source Bat Algorithm based Random Walk (PDD-MBARW), PDD with MS Simulated Annealing Random walker (PDD-MSSARW), PD-MRW, and PREDICT techniques. Primarily, obtain the gene set related to human HCC from CTD (version 2014). Afterwards, download gene expression profiles of HCC from Gene Expression Omnibus (GEO) <http://www.ncbi.nlm.nih.gov/geo/> datasets (GDS). The Drug Bank database combines by means of drug info with entire drug target info and Drug Bank database (version 4.0). In this research, we obtain the chemical-HCC associations from CTD (version 2014). This presented technique evaluates the consequence of various network aggregation methods on the whole ranking performance. Give five sets of results corresponding to five various cutoff values, that is to say, {1.25, 2.5, 5, 10, and 15}. The scores of the top-20 drugs are normalized with a logarithm so as to describe the results more obviously.

4.1 Performance analysis

Precision is defined as the ratio of correctly found positive observations to all of the expected positive observations.

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP}) \quad (20)$$

Sensitivity is defined as the ratio of correctly identified positive observations to the overall observations in the real class – yes.

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN}) \quad (21)$$

F1 score is defined as the weighted average of Precision as well as Recall. As a result, it produces false positives and false negatives.

$$\text{F1 Score} = 2 * (\text{Recall} * \text{Precision}) / (\text{Recall} + \text{Precision}) \quad (22)$$

Accuracy is calculated in terms of positives and negatives as follows:

$$\text{Accuracy} = (\text{TP} + \text{FP}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN}) \quad (23)$$

Here, by using a 5-fold cross-validation method reiterated 10 times, we evaluated the PDD-MSBARW method, existing PDD-MSSARW, PD-MRW, and PREDICT techniques. Table 1 represents the AUC and accuracy; recall averaged crosswise the Drug Bank classes with more than 20 drugs, using the GWIN as well as BGN networks. In Figure 2, the performance assessment results of the three various cutoffs of weight w values are calculated among five methods. Instead, the increased weight values, proposed PDDH-MICARW method provides improved precision value of 97.83% for w=15, when other methods such as PD-MSSARW, PD-MRW, and PREDICT methods provide precision value of 96.67%, 94.22% and 92.60% values respectively. The proposed technique performs superiorly when compared with other methods.

Table 1: Metrics comparison with Drug Bank categories with greater than 20 drugs

w cutoff values	Metrics	PREDICT	PD-MRW	PDD-MSSARW	PDD-MSBARW	PDDH-MICARW
w=5	Precision (%)	89.76	90.02	91.05	92.56	96.93
	Recall (%)	88.70	89.91	90.10	91.34	96.93
	F-measure (%)	89.23	89.965	90.575	91.95	96.93
	Accuracy (%)	88.35	89.71	91.82	92.76	94.40
w=10	Precision (%)	91.18	93.38	95.56	96.45	96.96
	Recall (%)	91.11	92.31	93.48	94.32	97.81
	F-measure (%)	91.145	92.845	94.52	95.385	97.38
	Accuracy (%)	91.25	92.49	93.53	94.65	95.20
w=15	Precision (%)	92.60	94.22	96.77	97.12	97.83
	Recall (%)	94.57	95.19	96.77	97.21	97.83
	F-measure (%)	93.585	94.705	96.77	97.165	97.83
	Accuracy (%)	92.64	93.41	94.75	95.68	96.00

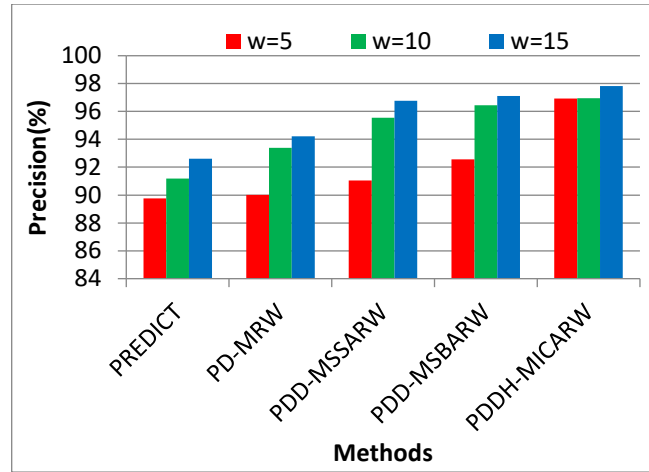


Figure 2: Precision vs. methods (with different weight values)

In Figure 2, the performance assessment results of the three various cutoffs of weight w values are calculated among five methods. Instead, the increased weight values, proposed PDDH-MICARW method provides improved precision value of 97.83% for $w=15$, when other methods such as PD-MSSARW, PD-MRW, and PREDICT methods provide precision value of 96.67%, 94.22% and 92.60% values respectively. The proposed technique performs superiorly when compared with other methods.

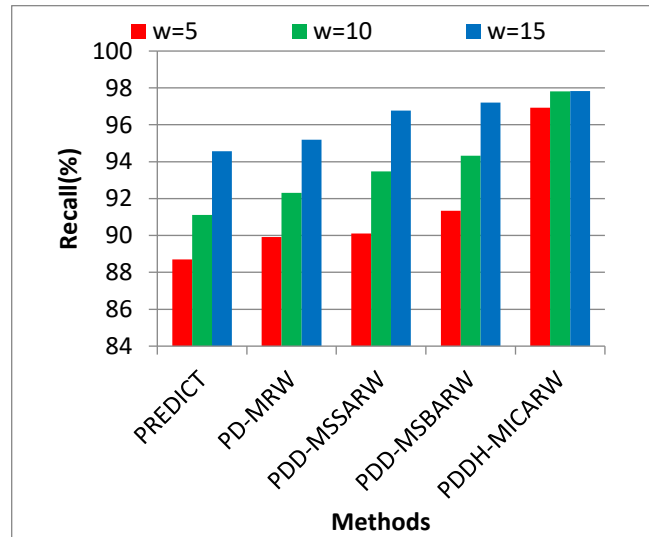


Figure 3: Recall vs. methods (with different weight values)

In Figure 3, the recall assessment results of the three various cutoffs of weight w values are calculated among five PDDH-MICARW methods, existing PDD-MSBARW, PDD-MSSARW, PD-MRW, and PREDICT techniques are represented. Instead, the increased weight values, proposed PDDH-MICARW technique provides a superior recall value of 97.83% for $w=15$, when other methods such as PDD-MSBARW, PDD-MSSARW, PD-MRW, and PREDICT techniques provide recall value of 91.21%, 96.77%, 95.19% and 94.57% values respectively. The proposed technique performs superiorly when compared with other methods.

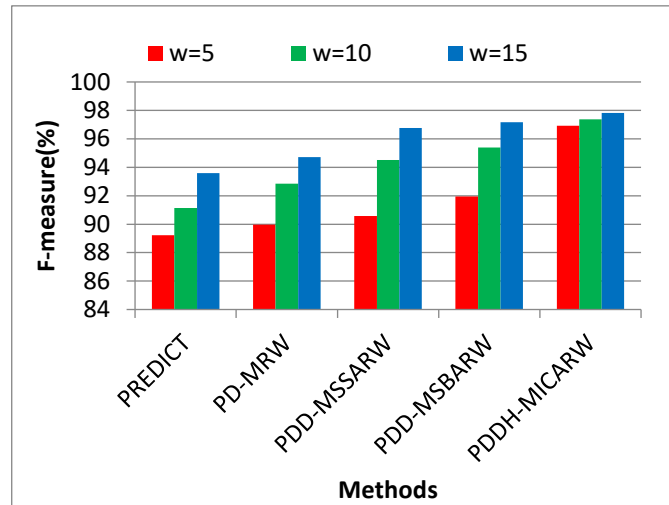


Figure 4: F-measure vs. methods (with different weight values)

In Figure 4, the f-measure assessment results of the three various cutoffs of weight w values are calculated among five PDDH-MICARW methods, existing PDD-MSBARW, PDD-MSSARW, PD-MRW, and PREDICT techniques are represented. Instead, the increased weight values, proposed PDDH-MICARW technique provides a superior f-measure value of 97.83% for $w=15$, when other methods such as PDD-MSBARW, PDD-MSSARW, PD-MRW, and PREDICT techniques provide f-measure value of 97.165%, 96.77%, 94.705% and 93.585% values respectively. The proposed technique performs superiorly when compared with other methods.

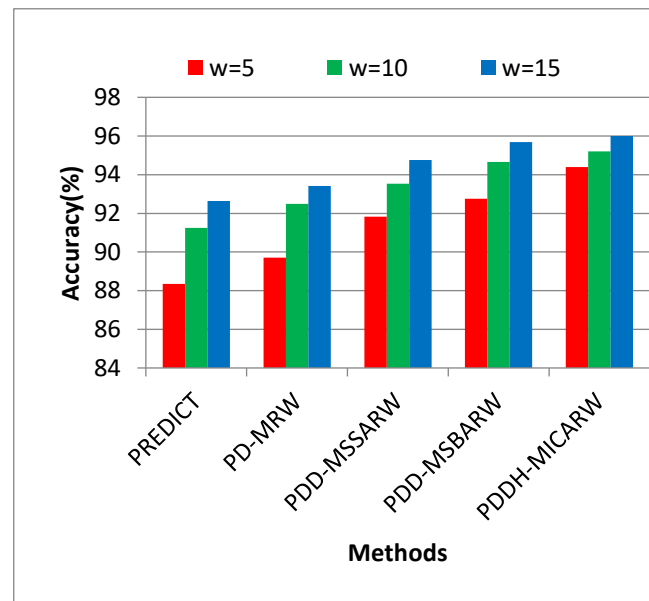


Figure 4: Accuracy vs. methods (with different weight values)

In Figure 5, the accuracy assessment results of the three various cutoffs of weight w values are calculated among five PDDH-MICARW methods, existing PDD-MSBARW, PDD-MSSARW, PD-MRW, and PREDICT techniques are represented. Instead, the increased weight values, proposed PDDH-MICARW technique provides a superior recall value of 96.00% for $w=15$, when other methods such as PDD-MSBARW, PDD-MSSARW, PD-MRW, and PREDICT techniques provide recall value of 95.68%,

94.75%, 93.41% and 92.64% values respectively. The proposed technique performs superiorly when compared with other methods.

5 Conclusion and Future Work

Drug repositioning is one of the most important methods for improving the drug discovery process. Most of the previous methods for drug repositioning don't focus on a specific disease and tend to use a specific source of data to predict drug-disease associations. In this study, focusing on a Hepatocellular Carcinoma (HCC), a new PDDH-MICARW method for novel drug usage prediction is proposed. PDDH-MICARW consists of several major steps: Firstly, integrate gene expression data, a gene set, and a protein-protein interaction (PPI) network to construct a gene-gene weighted interaction network for HCC (GWIN). Secondly, based on GWIN, use the theory of multi-source random walk to construct a drug-drug similarity network. Here, "multi-source random walk" means a random walk starting from multiple given vertices simultaneously, i.e., the multiple starting vertexes corresponding to multiple sources. A novel Improved Regularized Linear Discriminant Analysis (IRLDA) algorithm for dimensionality reduction is presented, which raised the prediction accuracy. Subsequently, a drug-drug similarity network is constructed, a disease-drug similarity network. Lastly, dependent upon the known drugs for HCC, score each drug in the drug-drug similarity network, disease-drug bipartite graph network. The proposed PDDH-MICARW method integrates heterogeneous data and knowledge, and uses the theory of multi-source random walk to obtain the similarity between each pair of drugs. In the simulation experiment, firstly, filter predicted drugs using the features of drug targets. The experimentation outcomes prove that the proposed method achieved greater performance outcomes in regard to accuracy, prediction, recall, and F-measure, matched up with existing techniques. Plan to broaden the applicability of our method by incorporating disease-specific symptoms and side effects of drugs are focused in future work. This is likely to be of significance in the near future, with the accumulation of clinical phenotypes and drug side effects, in conjunction with patient-specific genetic information.

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